Issues with Performance Pay: Risk Aversion and Multitasking

Group Discussion: Performance pay elicits effort but it has its limits

Example: Teachers

Teachers tend to dislike uncertainty over their compensation

Teachers' job requires effort on multiple tasks:

- Need to teach hard skills (spelling, high speed arithmetic)
- Need to teach soft skills (critical thinking, clear writing)

Form groups of 3-4

What do you think is the optimal bonus structure is for a teacher?

Outline

High-powered incentives impose risk

Multitasking

Discussion: Job flexibility

Key Points

Performance pay and Risk

High-powered incentives expose agent to risk

- Obtains high compensation if she is lucky and low if not

Last lecture model assumes that the agent is risk neutral: cares only about the expected value (mean) of her compensation

- Perhaps appropriate for CEOs
- Less appropriate for teachers, workers, bureaucrats

Most care about the mean and the variance of the compensation

- Are willing to earn less on average if the variance (uncertainty over how much they earn) is lower
- The higher the variance, the less happy they are

What happens when the agent is risk-averse?

Risk Aversion

Consider the following changes to the model:

- Output: $y = e + luck$ where $var(luck) = 1$
- Agent: chooses e to maximize $\mathbb{E}(w) - \gamma \cdot var(w) - c(e)$

Higher γ means higher risk aversion

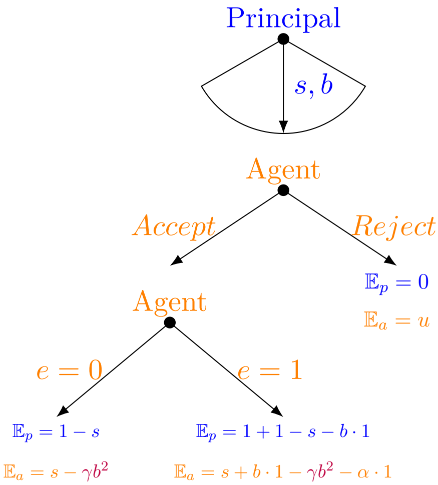
Math fact: $var(w) = var(s + b(e + luck)) = b^2$

Agent chooses e to maximize:

$$s + be - \gamma b^2 - \alpha e$$

Agent's problem is like before but her utility is smaller by γb^2

Game tree


Principal
s, b
Agent
Accept
Reject
$\mathbb{E}_p = 0$
$\mathbb{E}_a = u$
Agent
$e = 0$
$e = 1$
$\mathbb{E}_p = 1 - s$
$\mathbb{E}_a = s - \gamma b^2$
$\mathbb{E}_p = 1 + 1 - s - b \cdot 1$
$\mathbb{E}_a = s + b \cdot 1 - \gamma b^2 - \alpha \cdot 1$

Solving for optimal contract

Agent chooses e to maximize:

$$s + be - \gamma b^2 - \alpha e$$

Like before, chooses $e^* = 1$ only if $b \geq \alpha$

Hence Principal offers either $b^* = 0$ or $b^* = \alpha$

Agent's utility from $b^* = 0$ is as before so she demands $s^* = u$

Agent's utility from $b^* = \alpha$ is now

$$s + b - \gamma b^2 - \alpha$$

So if $b^* = \alpha$, she demands salary

$$s - \gamma b^2 = u \rightarrow s^* = u + \gamma b^2$$

So Principal's profit from contract $b^* = \alpha$ is

$$1 + \mathbb{E}(y - w) = 1 + e - s - be = 2 - u - \gamma\alpha^2 - \alpha$$

Optimal contract

Principal's profit if she offers $b^* = \alpha$ is

$$2 - u - \gamma\alpha^2 - \alpha$$

Recall that with $b^* = 0$ her profit is

$$1 - u$$

The former is smaller than the latter only if

$$1 - \gamma\alpha^2 - \alpha < 0$$

This will be satisfied if γ is sufficiently large

Offer no performance pay if agent is sufficiently risk averse

Optimal incentives and risk-aversion

Providing incentives to risk-averse agent is expensive

- Facing uncertainty over her pay she demands higher salary

A principal faces a tradeoff

- High bonus elicits effort but is costly
- Low bonus is cheaper but elicits less effort

In practice, high-powered incentives are more prevalent when workers are less risk-averse

- We see high-powered incentives for CEO's
- We see low-powered incentives for teachers

Retention Rule versus Performance Pay

Retention rule:

- Agent receives wage w only if her performance is above certain threshold $\bar{y}$
- Is fired and receives 0 otherwise

Similarly to performance pay, it is a risky contract

- It is expensive if agent is risk averse

There is no easy rule on whether retention rule or performance pay is better

Caveat: with retention contract, agent has incentive to *just meet the threshold*, which may distort effort

Performance Pay and Selection on Risk

Jobs offering high power incentives may attract risk neutral (or less risk averse agents)

Risk averse agents more likely to choose jobs with fixed wage

- Consider an agent with risk aversion γ'
- Her payoff (whether she works hard or not) is $s^* - \gamma' b^2 = u + \gamma b^2 - \gamma' b^2$
- Agents with low risk aversion γ' like this job more!

Changing compensation structure for existing workers should be done with care

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Multitasking

Most jobs consist of multiple tasks:

- Teachers need to teach hard skills (spelling, high speed arithmetic) and soft skills (critical thinking, clear writing)
- Police need to pursue offenders and minimize unnecessary confrontation
- Public utilities need to reduce costs and maintain high quality service

Performance on some tasks is easy to measure

Performance on other tasks is hard to measure

Multitasking in teaching

Consider a teacher who can exert effort on teaching

- Math skills e_m
- Critical thinking e_c

Suppose that intrinsically motivated teacher would spend x hours teaching, allocating this time as follows:

$$e_m = \frac{x}{2}, e_c = \frac{x}{2}$$

Suppose that students' math skills are measured by a test

$$y_m = e_m + luck$$

But students' critical thinking is not easily measurable

Should schools pay bonuses tied to performance on math test?

Multitasking in teaching

Bonuses tied to y_m can increase effort spent on math skills e_m:

$$e_m > \frac{x}{2}$$

What happens to e_c?

There is a limit on how much one can work:

- More time spent on e_m means less time spend on e_c

Incentivizing the effort one can measure disincentivizes the effort one cannot measure

If the latter is important, low-powered incentives are better

Example: No Child Left Behind

Quality of instruction is frequently topic of public policy

In 2002, president George W. Bush signed No Child Left Behind bill which tied school's compensation to students' performance on measurable tasks

President Obama on reforming No Child Left Behind:

> *Accountability is the right goal. Closing the achievement gap is the right goal. And we've got to stay focused on those goals. But experience has taught us that, in it's implementation, NCLB had some serious flaws that are hurting our children instead of helping them. Teachers too often are being forced to teach to the test. Subjects like history and science have been squeezed out.*

Examples

Financial crisis of 2009:

- Managers and traders at financial institutions were rewarded for short-term performance instead of long-term stability
- This contributed to catastrophic failures

British Petroleum:

- Provided high-powered incentives to refinery managers for short-term performance
- This led them to ignore long-term safety concerns
- Resulted in a catastrophic Deepwater Horizon oil spill in Gulf of Mexico in 2010

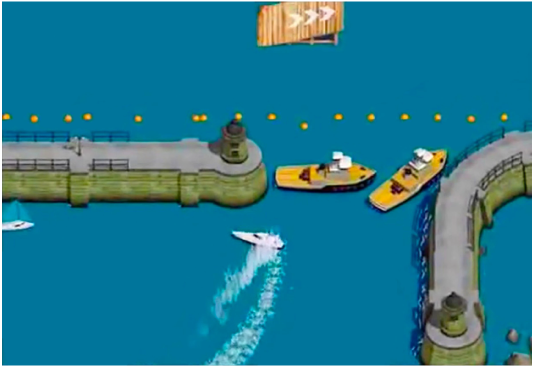
Multitasking and career concerns

Agents are frequently motivated by the next job

- They get a good job if they perform well in the current one
- If certain effort is more observable by outsiders than other, there will be effort distortions
- E.g., university administrators' fundraising is easily observable, their effort to raise instruction quality less so

Ideal incentive scheme would attach higher bonuses to tasks not easily observable by outsiders

Even AI makes the same mistake!



Incentive scheme needs to be tailored to specific situation

High powered incentives better when

- Performance on all effort easily measurable
- Effort measured precisely or agent not risk averse

Low powered incentives better when

- Performance on some effort not measurable
- Effort measured imprecisely or agent risk averse

Example: Apollo Program

Mission-driven objective: land a human on the Moon and return them safely

Multitasking: Tasks were complex, interdependent, and performance on many important tasks were difficult to measure individually

High risk: outcomes affected by factors beyond individual control

Implications:

- Strong incentives could crowd out intrinsic motivation, and cooperation
- Strong incentives on measurable outputs risked crowding out safety and quality

Example: Apollo Program

Predominantly fixed pay (Low-powered incentives)

- Engineers and scientists were largely salaried employees
- Compensation linked to specific performance goals (e.g., Grumman received bonuses for reducing the weight of the Lunar Excursion Module)

Reliance on:

- **Intrinsic motivation** and professional norms
- **Reputation** and **career concerns** rather than explicit bonuses

Example: CEO

Multiple tasks, but all contribute to profit

Profit is easy to measure

Position of CEO unlikely to attract risk averse individuals

Implications:

- Offer very high powered incentives
- CEOs are frequently rewarded with company shares

Caveat:

- Effort on maximizing short-term payoff are easy to measure
- Efforts maximizing long-term payoff are not
- CEOs compensated in shares may distort effort towards short-term goals

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Recent headlines on WFH

WORKFORCE

OPM directs agencies to quickly comply with Trump's return-to-office mandate

Agencies should aim for a 30-day deadline to implement Trump's return-to-office executive order, according to a memo from the Office of Personnel Management.

Drew Friedman | @dfriedmanWFED
January 23, 2025 2:13 pm 5 min read

Business Insider

The list of major companies requiring employees to return to the office, from JPMorgan and Dell to Amazon

Companies requiring workers to return to the office include AT&T, Amazon, JPMorgan, and Toyota. View a list of RTO mandates across business...

1 week ago

OPINION > TECHNOLOGY

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Biden's in-office work policy harmed federal employee retention. Trump's could destroy it.

BY GLEB TSIPURSKY, OPINION CONTRIBUTOR - 02/11/25 9:00 AM ET

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Discussion

Why would company want to offer work from home option?

Should this be combined with high-powered incentives?

Does this make more sense for JPMorgan or U.S. government to offer work from home option?

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Tailor the incentive scheme to the situation

Offer high-powered incentives when

- Intrinsic motivation unlikely
- Agent likely risk neutral
- Can measure performance on all of the important tasks